

Algebra
Applied Mathematics
Calculus and Analysis
Discrete Mathematics
Foundations of Mathematics
Geometry
History and Terminology
Number Theory
Probability and Statistics
Recreational Mathematics
Topology
Alphabetical Index
Interactive Entries
Random Entry
New in MathWorld
MathWorld Classroom
About MathWorld
Contribute to MathWorld
Send a Message to the Team
MathWorld Book

Wolfram Web Resources »

12,737 entries
Last updated: Fri Dec 18 2015

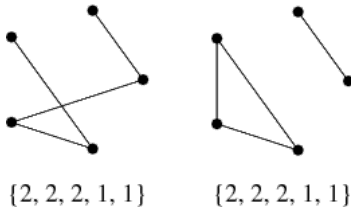
Created, developed, and
nurtured by Eric Weisstein
at Wolfram Research

Discrete Mathematics > Graph Theory > Graph Properties >
Interactive Entries > Interactive Demonstrations >

Degree Sequence

Given an **undirected graph**, a degree sequence is a monotonic nonincreasing sequence of the **vertex degrees** (valencies) of its **graph vertices**. The number of degree sequences for a graph of a given order is closely related to **graphical partitions**. The sum of the elements of a degree sequence of a graph is always even due to fact that each edge connects two vertices and is thus counted twice (Skiena 1990, p. 157).

The minimum vertex degree in a **graph** G is denoted $\delta(G)$, and the maximum degree is denoted $\Delta(G)$ (Skiena 1990, p. 157). A **graph** whose degree sequence contains only multiple copies of a single integer is called a **regular graph**. A graph corresponding to a given degree sequence can be constructed using `RealizeDegreeSequence[d]` in the **Wolfram Language** package `Combinatorica``.



It is possible for two topologically distinct graphs to have the same degree sequence. Moreover, two distinct convex polyhedra can even have the same degree sequence for their skeletons, as exemplified by the **triangular cupola** and **tridiminished icosahedron Johnson solids**, both of which have 8 faces, 9 vertices, 15 edges, and degree sequence $(3, 3, 3, 3, 3, 3, 4, 4, 4)$.

The number of distinct degree sequences for graphs of $n = 1, 2, \dots$ nodes are given by 1, 2, 4, 11, 31, 102, 342, 1213, 4361, ... (OEIS [A004251](#)), compared with the total number of nonisomorphic simple undirected graphs with n **graph vertices** of 1, 2, 4, 11, 34, 156, 1044, ... (OEIS [A000088](#)). The first order having fewer degree sequences than number of nonisomorphic graphs is therefore $n = 5$. For the graphs illustrated above, the degree sequences are given in the following table.

1	{0}
2	{0, 0}, {1, 1}
3	{0, 0, 0}, {1, 1, 0}, {2, 1, 1}, {2, 2, 2}
4	{0, 0, 0, 0}, {1, 1, 0, 0}, {2, 1, 1, 0}, {2, 2, 2, 0}, {3, 2, 2, 1}, {3, 3, 2, 2}, {3, 3, 3, 3}, {1, 1, 1, 1}, {2, 2, 1, 1}, {2, 2, 2, 2}, {3, 1, 1, 1}

The possible sums of elements for a degree sequence of order n are 0, 2, 4, 6, ..., $n(n - 1)$.

A degree sequence is said to be k -connected if there exists some k -connected graph corresponding to the degree sequence. For example, while the degree sequence $\{1, 2, 1\}$ is 1- but not 2-connected, $\{2, 2, 2\}$ is 2-connected.

SEE ALSO:
[Degree Set](#), [Graphic Sequence](#), [Graphical Partition](#), [k-Connected Graph](#), [Regular Graph](#), [Vertex Degree](#)

REFERENCES:
Ruskey, F. "Information on Degree Sequences." <http://www.theory.csc.uvic.ca/~cos/inf/nump/DegreeSequences.html>.
Ruskey, F.; Cohen, R.; Eades, P.; and Scott, A. "Alley CATs in Search of Good Homes." *Congres. Numer.* **102**, 97-110, 1994.
Skiena, S. "Realizing Degree Sequences." §4.4.2 in *Implementing Discrete Mathematics: Combinatorics and Graph Theory with Mathematica*. Reading, MA: Addison-Wesley, pp. 157-160, 1990.
Sloane, N. J. A. Sequence [A004251](#)/M1250 in "The On-Line Encyclopedia of Integer Sequences."

Referenced on Wolfram|Alpha: [Degree Sequence](#)

CITE THIS AS:
Weisstein, Eric W. "Degree Sequence." From *MathWorld*--A Wolfram Web Resource.
<http://mathworld.wolfram.com/DegreeSequence.html>

THINGS TO TRY:
= graph properties
= $(1, 1, -3)$ in spherical coordi
= domino tiling 4 steps

Interactive knowledge apps fr
Wolfram Demonstrations I

Average Vertex
of Connected C
John Cicilio

Fall in love with math all c
again...with Mathematica
Edition

Wolfram Web Resources

Mathematica »

The #1 tool for creating
Demonstrations and anything
technical.

Computerbasedmath.org »

Join the initiative for modernizing math
education.

Wolfram Problem Generator »

Unlimited random practice problems
and answers with built-in Step-by-
step solutions. Practice online or
make a printable study sheet.

Wolfram|Alpha »

Explore anything with the first
computational knowledge engine.

Online Integral Calculator »

Solve integrals with Wolfram|Alpha.

Wolfram Education Portal »

Collection of teaching and learning
tools built by Wolfram education
experts: dynamic textbook, lesson
plans, widgets, interactive
Demonstrations, and more.

Wolfram Demonstrations Project »

Explore thousands of free applications
across science, mathematics,
engineering, technology, business, art,
finance, social sciences, and more.

Step-by-step Solutions »

Walk through homework problems step-
by-step from beginning to end. Hints
help you try the next step on your own.

Wolfram Language »

Knowledge-based programming for
everyone.

 [Contact the MathWorld Team](#)

© 1999-2015 [Wolfram Research, Inc.](#) | [Terms of Use](#)